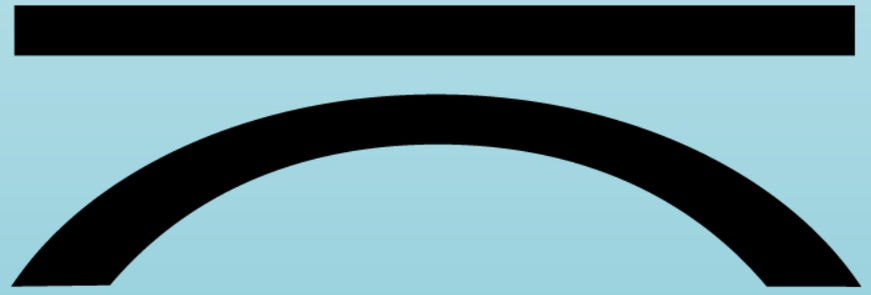
CISP 360

Introduction to Structured Programming

Dr. Fowler



FOLSOM LAKE COLLEGE
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Lightning Review

Lightning Review Questions

- What are the key elements of a generic if statement?
- What other questions to you have for me?

Code Debrief

coins.cpp

Code Debrief

Increases topic mastery.

Construct knowledge.

Programming Exercise

- Write a program for cyber league registration. There are 5 teams (1..5), but team 2 is full. Ask for the team the client wishes to join and if a valid choice say “Registered for team” x. If not a valid choice, provide as detailed a message as possible to the client and exit the program.

- Expect integer input, but don’t allow integers outside the team number (-0, 6+).

Debrief

What worked about this exercise?

What could be improved about this exercise?

Why did I ask you to do ….

Connect activity with course objectives/SLO's.

Connect activity with course readings.

The Point

- Chapter 4 is all about influencing **program flow** with decisions.
- These decisions are based on true or false **conditions**.

Chapter 5 - Looping

Agenda

5. Looping (Iteration)

    5.1. The While loop

    5.2. Input Validation with the While Loop

    5.3. Increment & Decrement Operators

    5.4. Counters

    5.5. Running Totals

    5.6. Sentinels

    5.7. Do .. While loop

    5.8. For loop

    5.9. Which Loop to use?

    5.10. Nested Loops

    5.11. Breaking out of a Loop

    5.12. Files for Data Storage

Agenda

5. Looping (Iteration)

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5.12. Files for Data Storage

What’s the big idea?

Repeating tasks over and over again is the secret of why we want computers to perform work.

Standardized.

Don’t get tired.

Same outcome.

- Loops have two parts:
  - a condition
  - plus, someplace to go when condition is true.

That sounds like an `if` statement …

it is like an if statement!

- The only difference between an if and a loop is **WHAT** happens to program control when it reaches the end of the loop block.
- If statements - execution follows main program flow **down** the page.
- Loop Statements - execution jumps **up** to the original loop statement.

Loop Elements

- **All** loops have these elements …
  - Loop Condition Location
  - Declare and Initialize Loop Control Variable
  - Loop when condition

Loop Condition Location

- Entry Condition Loop
- Exit Condition Loop

Loop Control Variables

- Inside the loop
- Everywhere else

Looping Condition

- Condition must evaluate to ?? for the loop to occur.

Loop Constructs

- While
- For
- Do/While

5.1 The While loop

```
statement1
while (test_expr)
      statement2
statement3
```

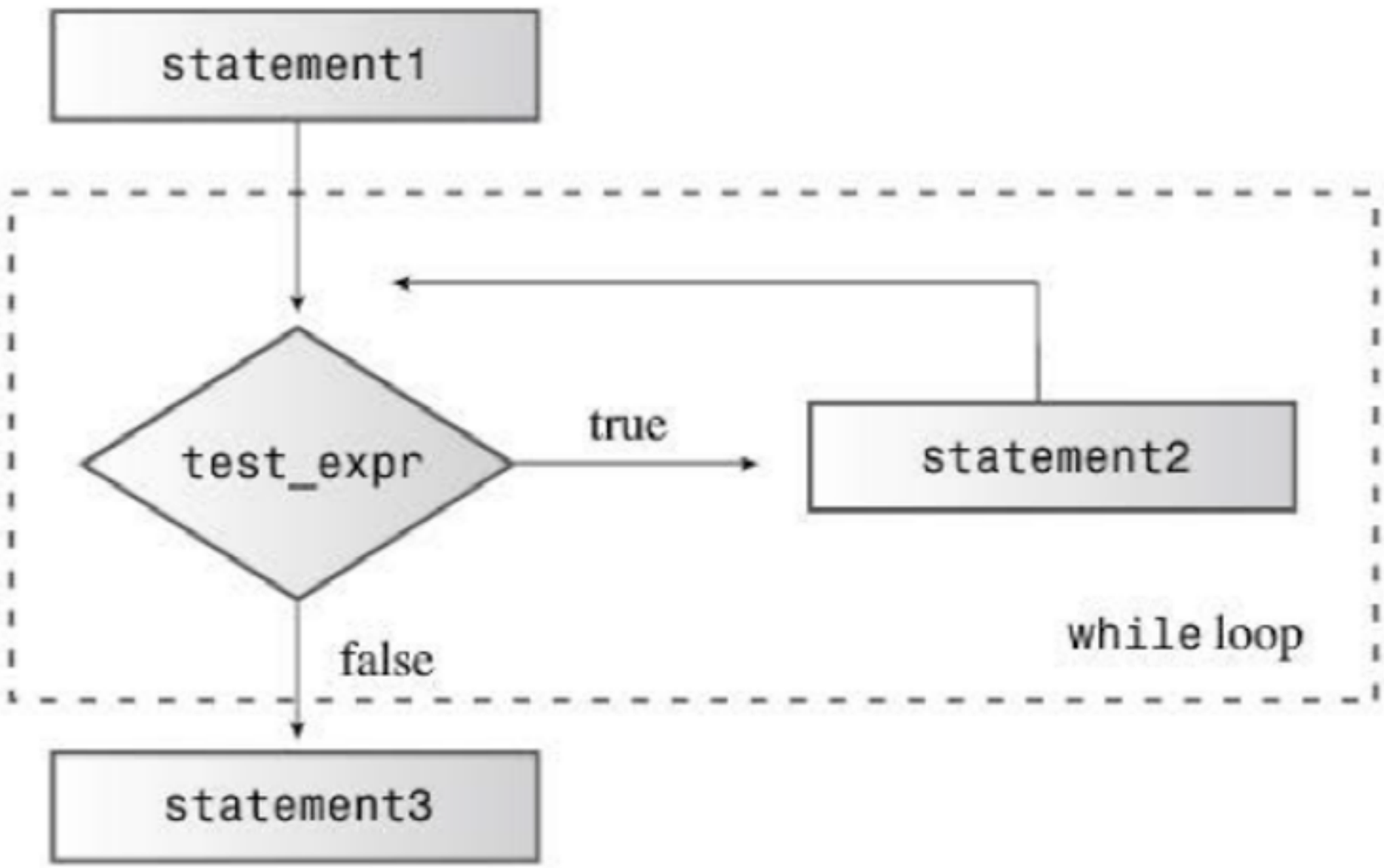

Figure 5.3 The structure of `while` loops.

Listing 5.13 **while.cpp**

```
// while.cpp -- introducing the while loop
#include <iostream>
const int ArSize = 20;
int main()
{
    using namespace std;
    char name[ArSize];

    cout << "Your first name, please: ";
    cin >> name;
    cout << "Here is your name, verticalized and ASCIIized:\n";
    int i = 0;                      // start at beginning of string
    while (name[i] != '\0')         // process to end of string
    {
        cout << name[i] << ": " << int(name[i]) << endl;
        i++;                        // don't forget this step
    }
    return 0;
}
```

```
// waiting.cpp -- using a clock in a time delayed loop
#include <iostream>
#include <ctime>    // the clock functions
int main()
{
    using namespace std;
    cout << "Enter the delay time, in seconds: ";
    float secs;
    cin >> secs;
    clock_t delay = secs * CLOCKS_PER_SEC;
    cout << "starting\a\n";
    clock_t start = clock();
    while (clock() - start < delay)
    ;
    cout << "done \a\n";
    return 0;
}
```

Desk Check

Exercise

Exercise Goals

- Review of code syntax
- Check understanding of syntax

Desk Check Exercise

- Pair up
- Manually determine the output of the following code?
  DO NOT TYPE THIS IN AND EXECUTE IT.
- We'll spend about 10 minutes on this, then go over it.

```cpp
// 0501Out.cpp -- What's the output?
#include <iostream>
using namespace std;

int main() {
    int i = 1;
    int num = 2;
    while (i < 5) {
        num = num * 1;
        i = i + 1;
    }
    cout << "i = " << i << " and num = "
        << num << endl;
}
```

```cpp
// 0501Out.cpp -- What's the output?
#include <iostream>
using namespace std;

int main() {
    int i = 1;
    int num = 2;
    while (i < 5) {
        num = num * 1;
        i = i + 1;
    }
    cout << "i = " << i << " and num = "
        << num << endl;
}
```

```
clang version 7.0.0-3
> clang++-7 -pthread
> ./main
i = 5 and num = 2
```

5.7 Do .. While loop

```
statement1
do
      statement2
while (test_expr);
statement3
```

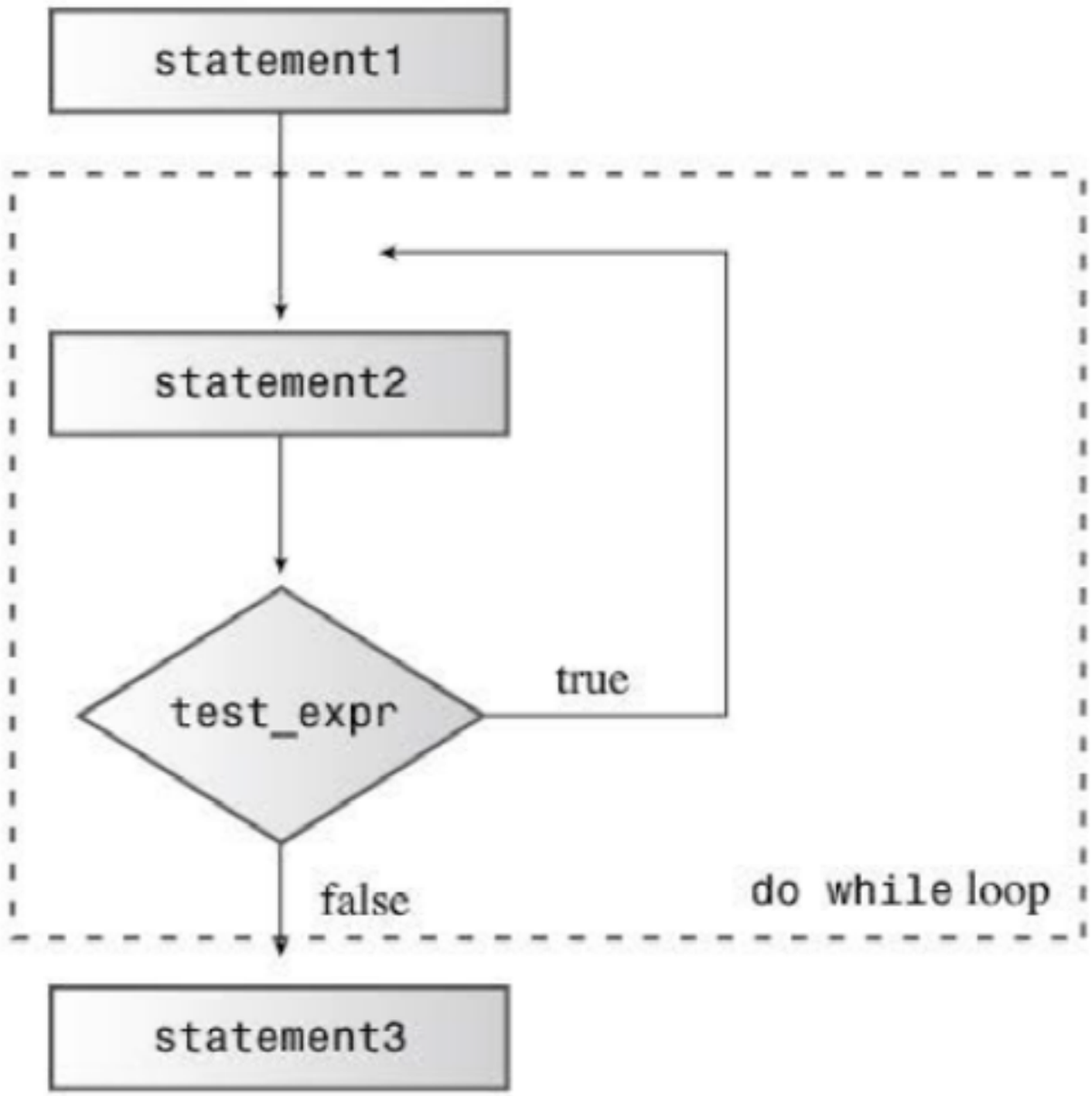

Figure 5.4 The structure of `do while` loops.

Listing 5.15 **dowhile.cpp**

```
// dowhile.cpp -- exit-condition loop
#include <iostream>
int main()
{
    using namespace std;
    int n;

    cout << "Enter numbers in the range 1-10 to find ";
    cout << "my favorite number\n";
    do
    {
        cin >> n;          // execute body
    } while (n != 7);      // then test
    cout << "Yes, 7 is my favorite.\n" ;
    return 0;
}
```

Exercise

Out0502.cpp

Exercise Goals

- Review of code syntax
- Check understanding of syntax

Desk Check Exercise

- Pair up
- Manually determine the output of the following code?
  DO NOT TYPE THIS IN AND EXECUTE IT.
- We'll spend about 10 minutes on this, then go over it.

```cpp
// 0520DoOut.cpp
// What's the output?
#include <iostream>
using namespace std;

int main() {
  int count = 0;
  int sum = 0;
  do {
    sum = sum + count * (count - 1);
    count++;
  } while (count <= 5);
  cout << sum << endl;
}
```

```cpp
// 0520DoOut.cpp
// What's the output?
#include <iostream>
using namespace std;

int main() {
  int count = 0;
  int sum = 0;
  do {
    sum = sum + count
    count++;
  } while (count <= 5)
  cout << sum << endl;
}
```

Output:

```
40
```

5.8 For loop

```
statement1
for (int_expr; test_expr; update_expr)
     statement2
statement3
```

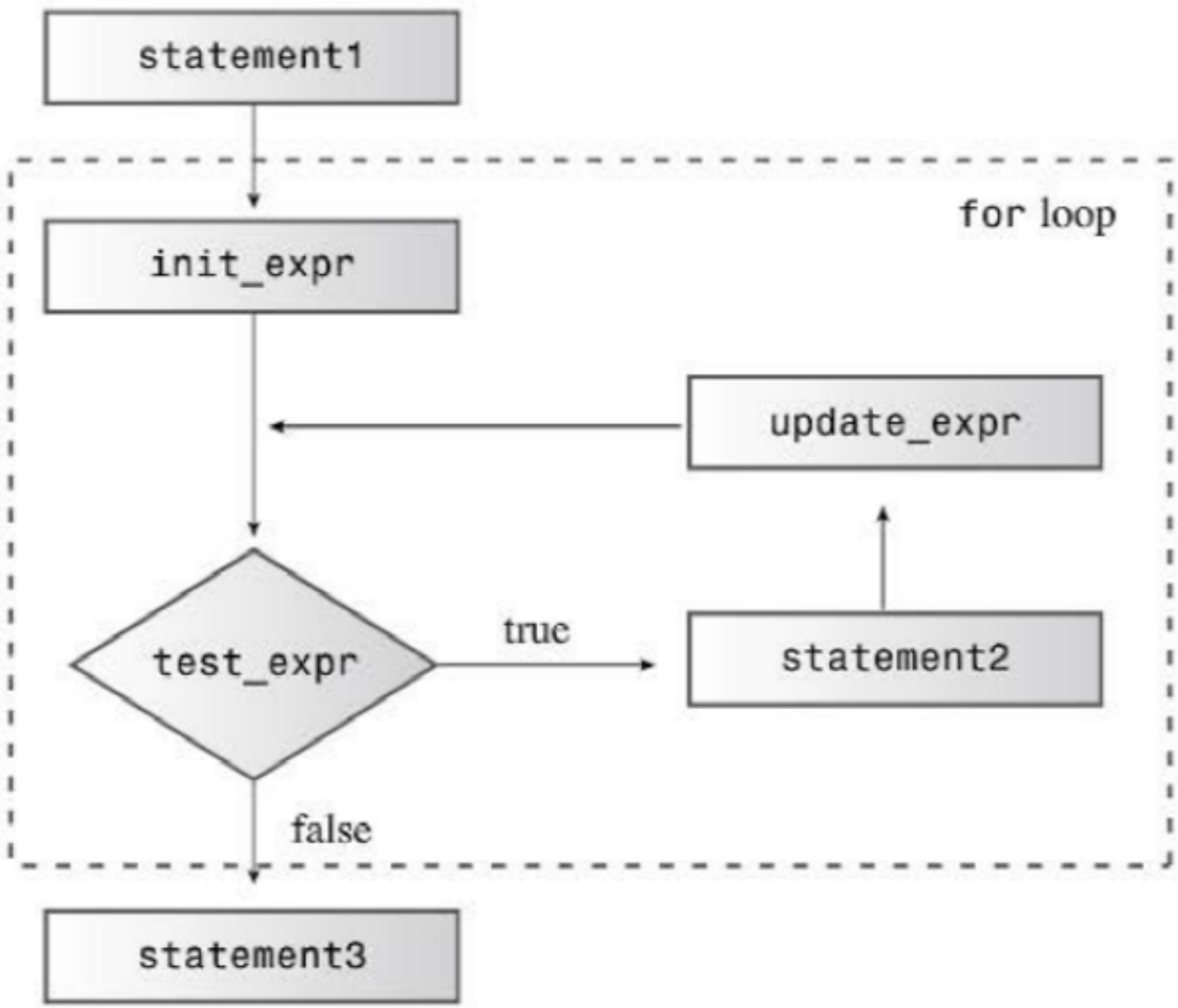

Figure 5.1 The design of `for` loops.

Listing 5.1 **forloop.cpp**

```
// forloop.cpp -- introducing the for loop
#include <iostream>
int main()
{
    using namespace std;
    int i;  // create a counter
//   initialize; test ; update
    for (i = 0; i < 5; i++)
        cout << "C++ knows loops.\n";
    cout << "C++ knows when to stop.\n";
    return 0;
}
```

Desk Check

Out0503.cpp

Exercise Goals

- Review of code syntax
- Check understanding of syntax

Desk Check Exercise

- Pair up
- Manually determine the output of the following code?
  DO NOT TYPE THIS IN AND EXECUTE IT.
- We'll spend about 10 minutes on this, then go over it.

```cpp
// 0530ForOut.cpp
// What's the output?
#include <iostream>
using namespace std;

int main() {
  for (int i = 0; i * i < 11; i++) {
    cout << i << " ";
  cout << endl;
  }
}
```

```cpp
// 0530ForOut.cpp
// What's the output?
#include <iostream>
using namespace std;

int main() {
  for (int i = 0; i * i < 11; i++) {
    cout << i << " ";
  cout << endl;
  }
}
```

Output:

```
0
1
2
3
```

5.9 Which Loop to use?

While Loop

Use when

- we don’t want to loop if the initial condition is false.

- we don’t know the number of times we will iterate.

do .. while Loop

Use when

- we want to loop at least once.

- good for menus and input validation.

for Loop

Use when

- we know how many times we need to loop.

Which loop to use?

You prompt the instructor for the number of students they will process test scores for. We want to enter 1 test score for each student in the class. We will not run the program if we have no students to process.

1) **while loop**
2) **for loop**
3) **do … while loop**
4) **other**

Which loop to use?

You prompt the instructor for the number of students they will process test scores for. We want to enter 1 test score for each student in the class. We will not run the program if we have no students to process.

1) while loop
2) **for loop**
3) do … while loop
4) other

We know the # of times to run.
We never skip the loop start.

Which loop to use?

You are reading record(s) in from the network and need to determine if age > 18. If yes, then a module is called for further processing. The record returns to main program flow when processing is complete. If no, the record continues with main program flow.

1) **while loop**
2) **for loop**
3) **do … while loop**
4) **other**

Which loop to use?

You are reading record(s) in from the network and need to determine if age > 18. If yes, then a module is called for further processing. The record returns to main program flow when processing is complete. If no, the record continues with main program flow.

1) while loop
2) for loop
3) do … while loop
4) **other**

There is no iteration as a part of the problem. Sure, it's implied, but it's not stated.

Which loop to use?

You are running a window manager which creates a main program window. You need to check for user input (mouse) and call the appropriate function(s). You want to continue checking until the quit button is pressed.

1) **while loop**
2) **for loop**
3) **do … while loop**
4) **other**

Which loop to use?

You are running a window manager which creates a main program window. You need to check for user input (mouse) and call the appropriate function(s). You want to continue checking until the quit button is pressed.

1) while loop
2) for loop
3) **do … while loop**
4) other

We always want to run through the code once.

Which loop to use?

You are reading a file where the first line is an integer count of the number of subsequent lines to process. You must read the line before you process the data to eliminate garbage data. You run this weekly and some weeks you have a count = 0.

1) **while loop**
2) **for loop**
3) **do … while loop**
4) **other**

Which loop to use?

You are reading a file where the first line is an integer count of the number of subsequent lines to process. You must read the line before you process the data to eliminate garbage data. You run this weekly and some weeks you have a count = 0.

1) **while loop**
2) **for loop**
3) do … while loop
4) **other**

Not #3 because if we have no records we will read garbage. So, we want to skip the loop body in that case.

Column Coding

Exercise

Shared Coding Exercise

Your row will be completing the same problem using the same style of loop. Once you are done, choose the best example(s) to share with the rest of the class.

You will share it by copying it and pasting it to a discussion group I will create.

Problem

Write a program which calculates the distance a vehicle travels (distance = speed * time). The client enters the speed and the total travel time. The program calculates and displays the distance traveled in 1 hour increments. Example: 3 hours at 40 mph yields

```
Hour              Distance
---               ---------
1                    40
2                    80
3                   120
```

Debrief

What worked about this exercise?

What could be improved about this exercise?

Why did I ask you to do ….

Connect activity with course objectives/SLO's.

Connect activity with course readings.

What's this section about?

Agenda

5. Looping (Iteration)

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What’s the big idea?

Repeating tasks over and over again is the reason why we want computers to perform work.

Standardized.

Don’t get tired.

Same outcome, every time.

NEXT TIME

Agenda

5. Looping (Iteration)

    5.1. The While loop

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Think-Pair-Present

Exercise

Think-Pair-Share

Purpose

Activate prior learning.

Promotes discussion and thought.

Answer any outstanding issues from last class.

Think

- Get some paper - not turn in.
- Take a few seconds to collect your thoughts.
- Think about the question I'm about to ask you.

Question

- What 3 words or phrases best sum up Chapter 4? Why

5

Pair

- Pair up
- Discuss your answers

5

Present

Jump to Presentation

Debrief

What worked about this exercise?

What could be improved about this exercise?

Homework Debrief

Exercise

Hw Debrief

Increases topic mastery.

Construct knowledge.

Process

- Pair up
- What did you find difficult for this assignment?
- What did you find interesting?

Hw3 Comments

- Compiler warnings by far largest error.
- Pretty good documentation of specifications
- Almost everybody's code ran
- Global variables rearing their head.
- Program greeting NOT the same as menu.

Coding Exercise

Write a program which prompts the client to enter the number of pennies, nickels, dimes and quarters they have. If the amount entered totals $1.00, congratulate them on winning. If the amount is more or less, tell them and also let them know how much under or over they are.

```cpp
// coin.cpp -- example in class activity
#include <iostream>
using namespace std;

// Function Prototype
void Prompt(string, float&);


int main() {
  float p,
        n,
        d,
        q,
        total;

  Prompt("pennies", p);
  Prompt("nickels", n);
  Prompt("dimes", d);
  Prompt("quarters", q);

  total = p + (n * 5.0) + (d * 10.0 ) + (q * 25.0);

  if (total > 100.0)
    cout << "You're over by " << total - 100.0 << "\u00A2" << endl;
    else if (total < 100.0)
      cout << "You're under by " << 100.0 - total << "\u00A2" << endl;
  else
    cout << "You win!" << endl;

return 0;
}

void Prompt(string str, float& f) {
  cout << "Enter the number of " <<str << ": ";
  cin >> f;
}
```

```cpp
// coin.cpp -- example in class activity
#include <iostream>
using namespace std;

// Function Prototype
void Prompt(string, float&);


int main() {
  float p,
        n,
        d,
        q,
        total;

  Prompt("pennies", p);
  Prompt("nickels", n);
  Prompt("dimes", d);
  Prompt("quarters", q);

  total = p + (n * 5.0) + (d * 10.0 ) + (q * 25.0);

  if (total > 100.0)
    cout << "You're over by " << total - 100.0 << "\u00A2" << endl;
    else if (total < 100.0)
      cout << "You're under by " << 100.0 - total << "\u00A2" << endl;
  else
    cout << "You win!" << endl;

return 0;
}

void Prompt(string str, float& f) {
  cout << "Enter the number of " <<str << ": ";
  cin >> f;
}
```

```
./main
Enter the number of pennies: 10
Enter the number of nickels: 4
Enter the number of dimes: 2
Enter the number of quarters: 3
You're over by 25¢
```